STOCHASTIC INVENTORY OPTIMIZATION UNDER UNCERTAIN DEMAND

Mini Project Report
Course: Stochastic Decision Making

Institution: SRM Institute of Science and Technology, Tiruchirappalli

Programme: B.Tech Artificial Intelligence

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Academic simulation note: All prices, demand scenarios, and probabilities used in this report are illustrative assumptions created for this mini project. They are not actual company records or a real demand forecast.

Abstract

Inventory decisions must often be made before customer demand is known. Purchasing too few units can cause lost sales, while purchasing too many can leave unsold stock. This project formulates a single-product inventory decision as a stochastic optimization problem. Three discrete demand scenarios are assigned probabilities, and candidate stocking quantities are evaluated using an expected-profit objective. A Monte Carlo simulation with 100,000 demand draws estimates profit variability and checks the exact expected-profit calculations. A deterministic baseline based on average demand is compared with the stochastic decision. Sensitivity analysis investigates how the optimal stock quantity changes when the probability of high demand changes.

1. Problem Statement

A retailer must choose a stock quantity before observing the actual demand for the selling period. Demand is uncertain and represented by a discrete probability distribution. The retailer earns revenue from units sold, receives a salvage value for each unsold unit, and pays the purchase cost for every unit stocked. The objective is to choose the stock quantity that maximizes expected profit.

Research question: Which stock quantity maximizes expected profit under the assumed demand distribution, and how does that decision compare with a deterministic decision based on average demand?

2. Objectives

• Define the decision variable, uncertainty, objective function, and constraints.

• Evaluate candidate stock quantities across all demand scenarios.

• Use Monte Carlo simulation to estimate expected profit and profit variability.

• Compare the stochastic decision with an average-demand baseline.

• Test sensitivity to changes in the probability of high demand.

3. Mathematical Formulation

3.1 Decision variable

Let Q be the number of units purchased before demand is observed. Q is restricted to non-negative quantities. For this implementation, the candidate set is $Q \in \{0, 5, 10, \ldots, 80\}$. The increment of five units is a modeling choice used to keep the decision search transparent.

3.2 Uncertain demand

Demand D (units)	Probability
20	0.20
40	0.50
60	0.30
Total	1.00

The probabilities sum to one. Expected demand is E[D] = 20(0.20) + 40(0.50) + 60(0.30) = 42 units.

3.3 Profit function

Let p = ■100 be the selling price, c = ■60 the purchase cost, and s = ■10 the salvage value of an unsold unit. For stock quantity Q and realized demand D, profit is:

$$\pi(Q,D) = p \times \min(Q,D) + s \times \max(Q-D,0) - cQ$$

The first term is sales revenue. The second term is the value recovered from unsold units. The last term is the total purchase cost.

3.4 Objective and constraints

Maximize expected profit: $\max_Q E[\pi(Q,D)] = \max_Q \Sigma_d P(D=d)\pi(Q,d)$.

Subject to $Q \in \{0,5,10,\ldots,80\}$. The decision is made before D is observed. The model assumes fixed prices and costs, one product, no capacity limit beyond the candidate range, and no replenishment during the period.

4. Methodology

The project uses a scenario-based stochastic optimization approach supported by Monte Carlo simulation. A small discrete demand distribution is suitable for a transparent educational model because every scenario can be enumerated and its contribution to expected profit checked directly.

Step 1 — Enumerate candidate decisions. Stock quantities from 0 to 80 units in increments of 5 are considered.

Step 2 — Calculate scenario profit. For each candidate quantity, calculate profit at demand levels 20, 40, and 60.

Step 3 — Calculate exact expected profit. Multiply each scenario profit by its probability and sum the results.

Step 4 — Run Monte Carlo simulation. Draw 100,000 demand outcomes using the stated probabilities and a fixed random seed. Calculate the profit associated with each candidate quantity for every simulated demand.

Step 5 — Select the best candidate. The quantity with the largest exact expected profit is selected. The Monte Carlo estimate is used as a numerical cross-check and to summarize profit variability.

Step 6 — Conduct sensitivity analysis. Keep the probability of medium demand at 0.50 and vary the probability of high demand between 0.10 and 0.50. The probability of low demand is adjusted so the total remains one.

Reproducibility: the Python code uses NumPy's random generator with seed 42, and saves results as CSV files and charts in the outputs directory.

Why use both exact enumeration and simulation?

Exact enumeration provides the expected value for the specified discrete distribution without sampling error. Monte Carlo simulation demonstrates how uncertainty can be evaluated computationally and estimates the distribution of realized profit. Comparing the two provides a useful correctness check.

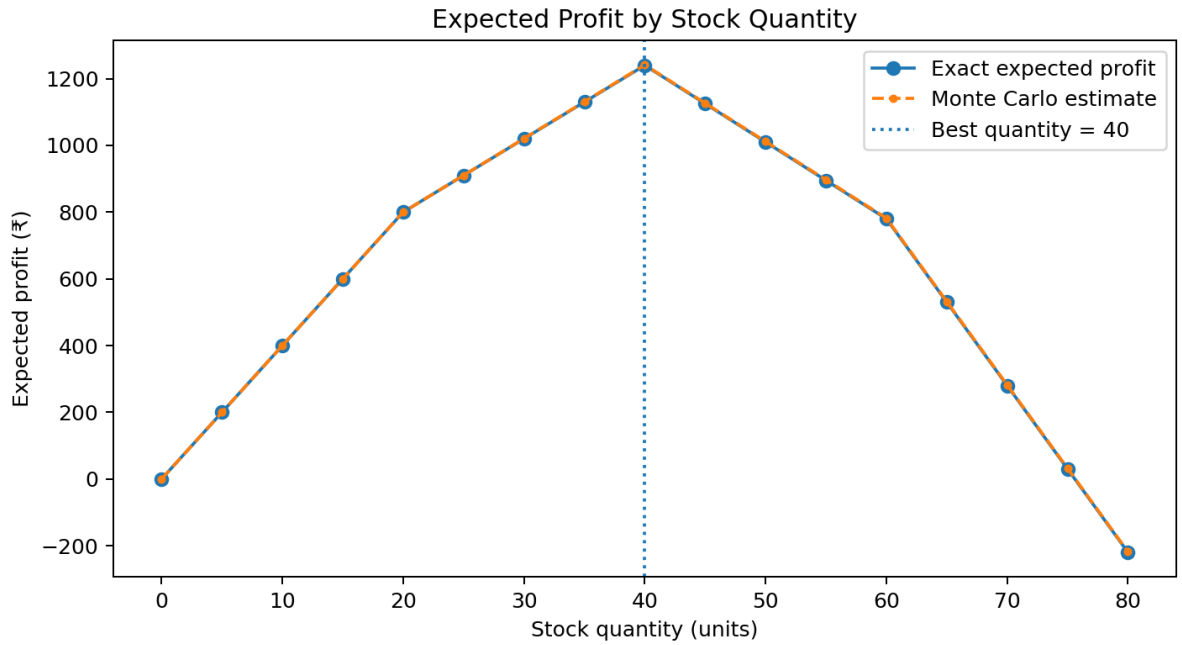
5. Results and Discussion

The best candidate quantity under the stated assumptions is **40 units**, with an exact expected profit of ■**1240.00**. The expected demand is 42.0 units. Because the candidate quantities use increments of five, the model searches a finite set of decisions rather than every possible integer quantity.

Selected candidate results

Stock Q	Profit if D=20	Profit if D=40	Profit if D=60	Expected profit
20	■800	■800	■800	■800.00
30	■300	■1200	■1200	■1020.00
35	■50	■1400	■1400	■1130.00
40	■-200	■1600	■1600	■1240.00
45	■-450	■1350	■1800	■1125.00
50	■-700	■1100	■2000	■1010.00

The table illustrates that a stocking decision should be judged across all plausible demand outcomes, not only one outcome. The expected-profit calculation weights each outcome by its assigned probability.



The Monte Carlo estimate at the selected quantity is ■1241.42. It should be close to the exact expected profit; small differences arise because simulation samples a finite number of random demand outcomes.

6. Deterministic Baseline and Sensitivity Analysis

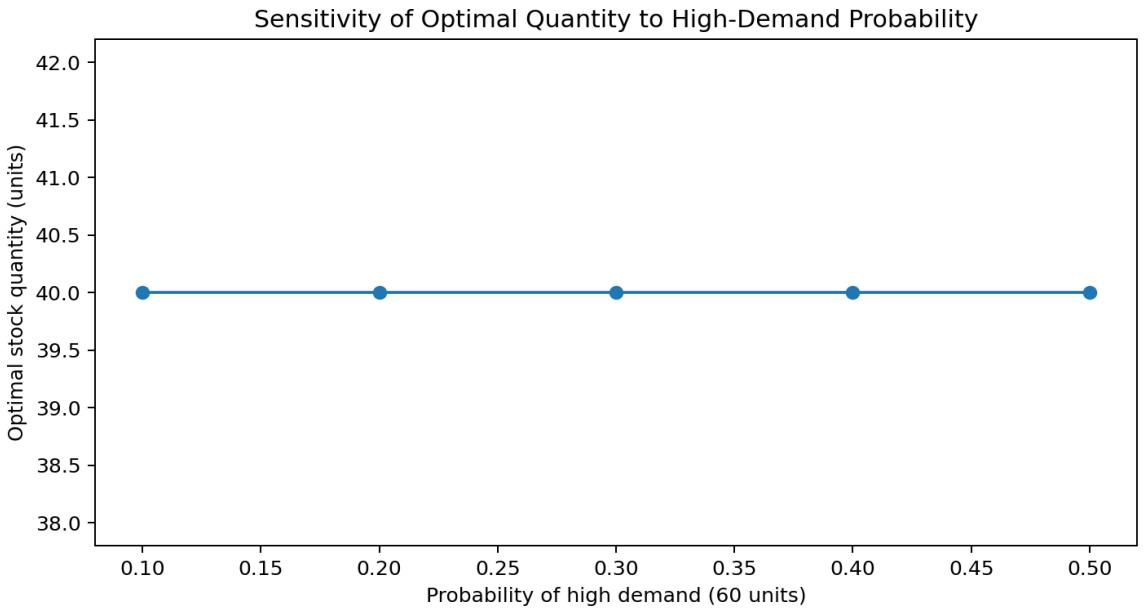
6.1 Deterministic baseline

The deterministic baseline uses the expected demand (42.0 units) as though it were known with certainty. The nearest candidate quantity is 40 units. Evaluated under the same uncertain demand distribution, this quantity gives an expected profit of ■1240.00. The stochastic search evaluates all candidate quantities and chooses the one with the highest expected profit. Under the current illustrative assumptions, the comparison is the same quantity; this is a useful result rather than a failure, because it shows that a stochastic method does not always have to produce a different decision. The value comes from explicitly accounting for probabilities and scenario outcomes.

6.2 Sensitivity analysis

The high-demand probability is varied while the medium-demand probability remains at 0.50. The low-demand probability is the remaining probability, ensuring the distribution sums to one.

P(D=20)	P(D=40)	P(D=60)	Optimal Q	Max expected profit
0.40	0.50	0.10	40	■880.00
0.30	0.50	0.20	40	■1060.00
0.20	0.50	0.30	40	■1240.00
0.10	0.50	0.40	40	■1420.00
0.00	0.50	0.50	40	■1600.00

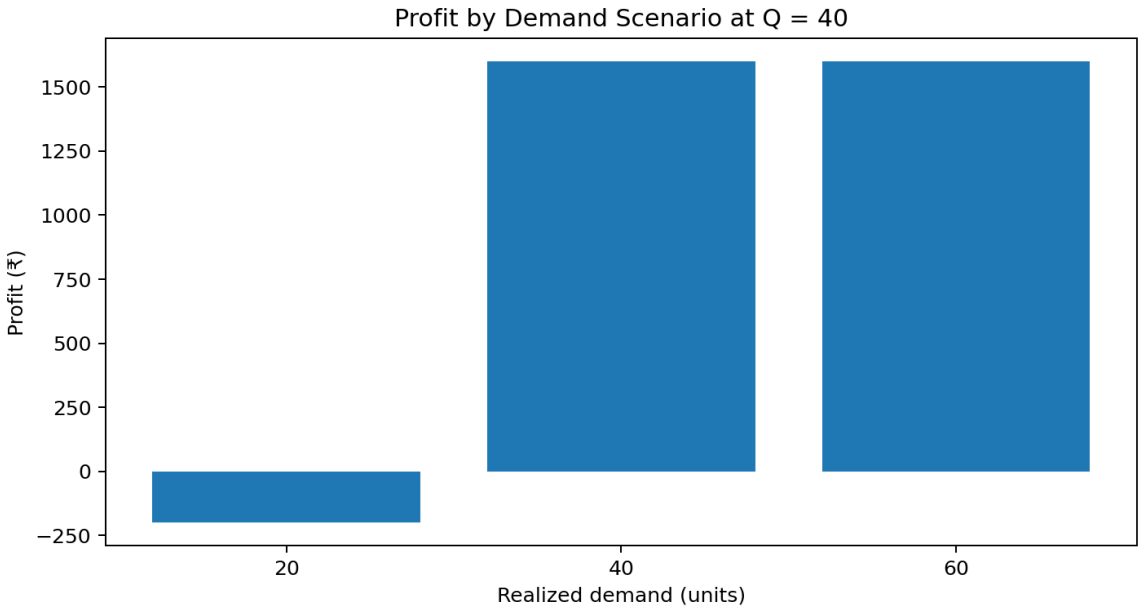


Sensitivity results indicate how the preferred quantity responds to a changed probability distribution. The result is conditional on the assumed selling price, purchase cost, salvage value, and demand scenarios.

7. Risk, Limitations, and Conclusion

7.1 Profit variability

Expected profit summarizes average performance but does not describe every possible outcome. The simulation also records the standard deviation and 5th and 95th percentiles of profit for each candidate quantity. These metrics provide additional information about variability and downside or upside outcomes. They are descriptive statistics under the assumed distribution, not guarantees of future profit.



7.2 Limitations

• Demand is represented by only three possible values and probabilities are assumed rather than estimated from historical sales.

• Selling price, purchase cost, and salvage value are fixed for all scenarios.

• The model includes one product and one selling period; it does not model replenishment, lead time, storage limits, spoilage, or supplier constraints.

• Candidate stock quantities are restricted to multiples of five between 0 and 80 units.

• Results should not be generalized to a real retailer without validated data and additional constraints.

7.3 Conclusion

This project formulated an inventory decision as a stochastic optimization model and evaluated candidate stock quantities using expected profit. Monte Carlo simulation provided a computational estimate of expected profit and profit variability, while exact scenario calculations provided a benchmark. The selected quantity under the illustrative assumptions is 40 units. Sensitivity analysis demonstrated how a change in the demand distribution can influence the decision. The main conclusion is that explicit uncertainty modeling offers a structured way to compare inventory choices, although the quality of the recommendation depends on the realism of the assumptions.

References

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5. Course assignment handout: Mini Project — Stochastic Optimization Model, Stochastic Decision Making.